

Arpit Mathur

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PROFILE

I'm a Ph.D. student in Human-Computer Interaction at Carnegie Mellon University focused on AI-driven healthcare visualization and decision support. My research integrates machine learning, human-AI collaboration, and data visualization to design interpretable, workflow-centered tools that enhance understanding and shared decision-making in clinical care.

SKILLS

Research Methods: Systematic Literature Reviews, Ethnographic Observation, Contextual Inquiry, Think-Aloud Studies, Semi-Structured Interviews, Self-Assessment Surveys, Thematic Analysis

Programming Languages: Python, Javascript, Java, C++, C#

Techniques: Deep Learning, Data Visualization, NLP, UI Design, Computer Graphics

Tools: Adobe Creative Suite, Sketch, Figma, Invision, Tableau, Matlab

EDUCATION

Carnegie Mellon University

Ph.D. - Human-Computer Interaction

Advisor: Adam Perer

Pittsburgh, PA

Aug 2024 - Present

Georgia Institute of Technology

Master of Science - Human-Computer Interaction

Atlanta, GA

Aug 2019 - May 2021

Indian Institute of Technology, Guwahati

Bachelor of Design - Design

Minor in Mathematics

Guwahati, India

Aug 2015 - May 2019

PUBLICATIONS

Under Review

Arpit Mathur, Katelyn Morrison, Udhirna Krishnamurthy, Jonathan Lai, Sang Wan Lee, Priscilla Correa-Jaque, Shili Lin, Sandeep Sahay, Megan Griffiths, Manreet Kanwar, Allen Everett, Raymond Benza, and Adam Perer. Progressive Disease Patient Perspectives on AI Survival Assessments: Current Practices, Barriers, and Opportunities, Under Review, 2025.

W. Epperson, **A. Mathur**, D. Moritz, and A. Perer, [Texture: Structured exploration of text datasets](#), Under Review, 2025.

Conference Papers

Morrison, K., **Mathur, A.**, Bradshaw, A., Wartmann, T., Lundt, S., Zandifar, A., Dai, W., Batmanghelich, K., Eslami, M., & Perer, A. (2025). [A Human-Centered Approach to Identifying Promises, Risks, & Challenges of Text-to-Image Generative AI in Radiology](#). Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, 8(2), 1758-1770.

Ben Rydal Shapiro, Rogers Hall, **Arpit Mathur**, and Edwin Zhao. 2025. [Exploratory Visual Analysis of Transcripts for Interaction Analysis in Human-Computer Interaction](#). In CHI Conference on Human Factors in Computing Systems (CHI '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 17 pages.

Bradshaw, A., Morrison, K., **Mathur, A.**, Dai, W., Eslami, M., Batmanghelich, K., & Perer, A. (2025). [Toward Interpretable 3D Diffusion in Radiology: Token-Wise Attribution for Text-to-CT Synthesis](#). In Medical Imaging with Deep Learning-Short Papers.

Shapiro, B. R., Hall, R., **Mathur, A.**, & Zhao, E. (2024). [Turn Charts for Interaction Analysis: Visually Mapping the Conversation Floor](#). In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024 (pp. 43-50). International Society of the Learning Sciences.

Mathur, A. & Shapiro, B.R. (2022). [Interactive Transcription Techniques for Interaction Analysis](#). In Proceedings of the 16th International Conference of the Learning Sciences (ICLS). International Society of the Learning Sciences, pg. 19-26. Hiroshima, Japan: International Society of the Learning Sciences.

Fulfagar, L., Gupta, A., **Mathur, A.**, Shrivastava, A. (2021). [Development and Evaluation of Usability Heuristics for Voice User Interfaces](#). In: Chakrabarti, A., Poovaiah, R., Bokil, P., Kant, V. (eds) Design for Tomorrow—Volume 1. ICoRD 2021. Smart Innovation, Systems and Technologies, vol 221. Springer, Singapore.

Mathur, A., Bhowmick, S., Sorathia, K. (2019). [A Study of Outbound Automated Call Preferences for DOTS Adherence in Rural India](#). In: Lamas, D., Loizides, F., Nacke, L., Petrie, H., Winckler, M., Zaphiris, P. (eds) Human-Computer Interaction – INTERACT 2019. INTERACT 2019. Lecture Notes in Computer Science(), vol 11748. Springer, Cham.

Workshop Papers

Mathur, A., Perer, A. (2025). [Clini-Compare: An Interactive Patient-Similarity Visualization Tool for Clinical Decision Support](#). In: Avellino, I., Kuo, P.-Y., Foong, P. S., Wiese, J., Mentis, H., Munson, S., Wallace, J., Singh, A., Miller, A., Epstein, D., & Nunes, F. (2025, April 12). CHI '25 Workshop on Envisioning the Future of Interactive Health.

Poster Presentations

Mathur, A., Bruce, C. (2021). [A Study of Motivation, Preferences, and Pain Points Regarding Participation in Career Related Mentorship](#). In: Stephanidis, C., Antona, M., Ntoa, S. (eds) HCI International 2021 - Posters. HCII 2021. Communications in Computer and Information Science, vol 1419. Springer, Cham.

INDUSTRY EXPERIENCE

The MathWorks

Natick, MA

User Experience Designer

Jun 2021 - Jun 2024

- Supported multiple MATLAB and Simulink products, including MATLAB fixed-point workflows, Simulink Studio, and System Composer.
- Led 10+ collaborative design events, developed wireframes, and iteratively prototyped high-fidelity designs using Figma.

SAS Research and Development

Pune, India

User Experience Design Intern

May 2018 - Jul 2018

- Redesigned the SAS Analytics Credit Scoring Tool. Conducted Contextual Interviews with Risk Analysts in Indian banks, and used those insights to create high-fidelity sketch mockups.

RESEARCH EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Doctoral Research Assistant

Aug 2024 - Present

- Conducting research at the intersection of Human-Computer Interaction, Machine Learning, Healthcare, and Human-AI Collaboration, focusing on developing innovative solutions in healthcare using advanced data visualization techniques.

Georgia Institute of Technology

Atlanta, GA

Graduate Research Assistant

Jan 2020 - May 2021

- Designed and Developed Interactive web based visualizations as course material for EC 3040 Semiconductor Devices.
- 4 interactive chapters were used by 100 students enrolled in EC 3040 over the fall semester 2020, to facilitate remote learning.

Indian Institute of Technology, Guwahati

Guwahati, India

Undergraduate Research Assistant.

Aug 2018 - May 2019

- Worked with the Govt. of Assam to Design and Test an IVR based Information Dissemination solution in 3 Rural districts of Assam, with 2000+ Patients.
- The IVR system was used to measure and improve adherence to the Tuberculosis (TB) Directly Observed Therapy (DOTS).

Usability Engineering and Human-Computer Interaction Lab, IITG

Guwahati, India

Undergraduate Research Intern.

Aug 2017 - Dec 2017

- Designed an iOS app for health and battery management of electric vehicles.
- Used Surveys and Semi-Structured Interviews for data collection
- Conceptualized User Flows, Low and High-Fidelity mockups for the app

TEACHING EXPERIENCE

The MathWorks

Atlanta, GA

Lead Instructor

Aug 2023

- Led the Try Design Workshop, a two day event to teach design methods to a multi-disciplinary group of contributors.

Georgia Institute of Technology

Atlanta, GA

Graduate Teaching Assistant

Aug 2020 - Dec 2020

- Facilitated remote learning by leading online lab sessions, virtual office hours, and offering project support via discussion boards and video calls.

- Developed additional online resources and recorded tutorials to help students with d3.js visualizations in a fully remote setting.

Georgia Institute of Technology

Atlanta, GA

*Graduate Teaching Assistant.**Jan 2020 - May 2020*

- Provided in-person support, guiding students through projects using d3.js and assisting with lab sessions, office hours, and project evaluations.
- Collaborated with the instructor to create course materials and grading rubrics, ensuring fair and consistent assessments.

HONORS AND AWARDS

CMLH Generative AI in Healthcare Fellowship, Carnegie Mellon University (2024)

Awarded a \$100,000 fellowship supporting research on generative AI applications in healthcare and human-centered design.

HackGT Winner (2020)

Ranked among the top 8 out of 1,500 teams in HackGT 7: Reimagine Reality.

ICoRD'21 2021 - Most Distinguished Paper

Development & Evaluation of Usability Heuristics for Voice User Interfaces.

IndiaHCI 2018 - Winner, Student Design Consortium

First prize at the IndiaHCI 2018 Student Design Consortium.

EXTRACURRICULAR

Volunteer - Guwahati Half Marathon (2017, 2018)

Helped Organize the Guwahati Half Marathon - Run for a Tobacco Free India, as a student volunteer.

Secretary - Coding Club IIT Guwahati (2018)

Led the student organized coding club. Helped conduct hackathons, student lectures, and projects for the student community.

Head Organizer - TechExpo (2018)

TechExpo 2018 - The largest student organized technical competition in India.